CSI 4160 M1 Report: Quikvote

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Introduction

In just the two weeks leading up to this report, our group has had four discussions/polls to reach consensus: choosing the best time for the biweekly meetings with the class TA, choosing a project idea, choosing a tech stack, and choosing a project name. Each of these decisions require several days of back and forth due to the asynchronous nature of online messaging, and due to the desire to be fair and not step on each others' toes. Everybody has experienced at some point the indecisiveness of a group on where to eat, what to do, what movie to watch, and more. Quikvote aims to solve this by providing a zero friction platform to quickly create polls and get consensus from your group, using a variety of techniques and tools to simplify the poll creation process as well as the vote reconciliation process.

Motivation

Polls are used everywhere by everyone today, and yet the existing platforms for polls are lackluster, due to the perceived simplicity of polls. But none of them nail down on the dynamics of polling and vote reconciliation. For example, one of the most popular poll platforms Strawpoll strictly counts votes. But there are several aspects of voting left to be desired. Some voters feel very strong about their choice, while other voters do not, and yet their votes have the same weight. Some voters feel very strongly opposed to the result and would like to veto the result, but these poll platforms have no such notion.

The goal of Quikvote is to help people overcome analysis paralysis, saving their limited time and resolving conflict quicker and easier. The concept of polls can be applied to groups of any size: a couple, a group of friends, a conference, a school, an organization, etc. Simplifying this process at a granular level has the potential to massively scale up and have exponentially beneficial effects on individuals.

Problem Statement and Scope

The problem Quikvote addresses is the friction and inefficiency inherent in group decision making, and the lack of flexibility in existing solutions. Existing platforms fall into two buckets: simple polling tools that only count votes, and complex survey platforms that are overkill for informal decisions. And, neither category addresses the nuanced dynamics of real-world group consensus building, such as varying levels of voter conviction, the need for tie-breaking mechanisms, and the challenge of generating options in the first place.

The scope of this project includes the development of a web-based platform with the following capabilities:

1. Instant poll creation and voting without requiring user accounts
2. Poll sharing (public or password-protected) to participants
3. Support for weighted voting where users can express varying levels of conviction
4. Integration with Large Language Models (LLMs) for intelligent option generation
5. Randomness-based tie-breaking
6. Notification services to alert participants of poll updates
7. Cloud infrastructure for hosting and data persistence.

Some other features remain out of scope, including user profiles, historical analytics, and complex authentication systems. The focus remains on the core polling experience enhanced by strategic API integrations that provide genuine value for spontaneous group decisions.

Literature Review

Research in social choice theory and group decision making provides valuable context for the challenges Quikvote aims to solve. Zou et al. (2015) analyzed over 340,000 Doodle polls to study strategic voting behavior in approval voting systems. Their findings revealed that participants in open polls, where votes are visible, exhibit significantly different behavior compared to hidden polls, with voters more likely to approve popular or unpopular options while avoiding intermediate choices. These results are not shocking but emphasize the importance of anonymous voting for ensuring honesty. The moment one of the members of the group states their preferences, all the other members are biased and the results will be skewed.

Freixas and Pons (2021) examined the mathematical properties of anonymous and weighted voting systems, demonstrating the complexity of designing fair aggregation mechanisms when voters have multiple options and outcomes are multidimensional. Their work is particularly relevant to our goal in creating Quikvote of supporting weighted preferences, where not all votes carry equal intensity. This study pioneered the research of niche voting dynamics, and the fact that the study was published in 2021, indicates that there is a clear gap in the market that can leverage our modern knowledge of human psychology in getting consensus between groups, which is the slot Quikvote aims to fill.

Finally, Huang et al. (2024) developed a collaborative group decision support system using multi-actor, multicriteria analysis (MAMCA), demonstrating how structured approaches can hugely improve group decision outcomes. Their work supports our approach to making the voting and decision process more systematic without adding significant complexity. Additionally, this study's recent publication date suggests even further that this is an evolving field; it is not a "solved" problem. We know it is not a solved problem because 99%+ of people do not use any kind of polling tool for informal use cases, which we hope to change.

Gap Identification

Despite extensive research on voting theory and the availability of various polling tools, a significant gap exists between academic understanding and practical implementation for casual group decision-making. Existing commercial solutions treat voting as a simple counting exercise, ignoring the nuanced dynamics that research has shown to be significant for maximizing fairness and voting effectiveness.

The major gap that exists in the treatment of vote intensity. Current platforms give equal weight to every vote, but Freixas and Pons's work on weighted voting systems shows that this is not always appropriate. A person who feels strongly about a choice should arguably have more influence than someone who is indifferent. Quikvote bridges this gap by allowing users to express different levels of conviction.

One final gap we aim to fill is the existence of tie-breaking mechanisms. Existing solutions gather results and declare a winner, wiping their hands and claiming that their job is done. But our research has shown that the initial batch of results is just the beginning of an effective polling loop. Seeing the results of a poll trigger some voters to second guess themselves. Some voters have egregious reactions to the result and try to veto the result, which would typically cause conflict in a group setting, but this is a keystone feature of Quikvote.

Proposed System

Quikvote is designed as a full-stack microservices-based application with the following architecture:

- Quikvote Website:
  - Technologies used: TypeScript, React, Vite, TanStack
  - The website is the interface for users to interact with the platform. It is best fitted to be a web application due to its innate ability for users to access the platform from any device on any operating system without needing to commit to installing anything
- Voting Service (Custom REST API & BFF API):
  - Technologies used: Python, Fast API, Magnum, AWS Lambda, API Gateway
  - This custom-built service handles all the core domains: poll creation, vote submission with optional weighting, and result aggregation. It maintains the core business logic and data models for polls. It also acts as the BFF (backend-for-frontend) for the website
- Notification Service:
  - Technologies used: Twilio for SMS, SendGrid for email
  - Integrates with notification APIs to alert participants when polls are created, updated, or closed. This service operates asynchronously to avoid blocking the main voting flow
- Suggestion Service:
  - Technologies used: Google Gemini (or any other major LLM)
  - Interfaces with LLM APIs to generate intelligent option suggestions based on context provided by the poll creator (e.g., "Italian restaurants near downtown" or "team-building activities for 10 people")
- Randomness Service:
  - Technologies used: Random.org

    - Integrates with randomness APIs to provide fair tie-breaking when polls result in exact ties, adding an element of chance that can help groups move past indecision.
- Infrastructure:
    - Technologies: AWS, AWS CDK
    - The entire system will be deployed on AWS using services including Lambda, API Gateway, S3, and more, leveraging the CDK for declarative IaC to achieve stable CI/CD

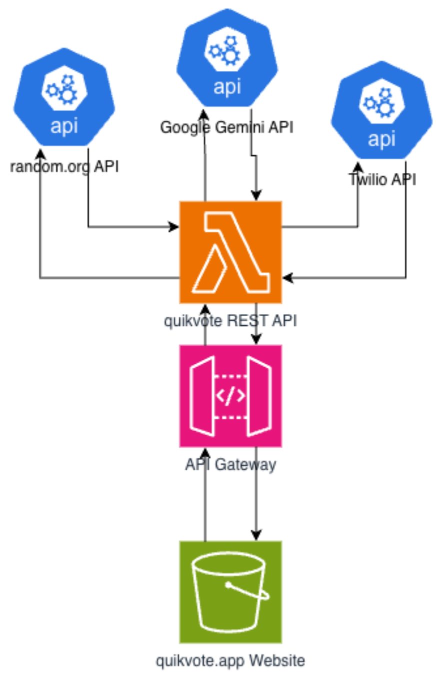

Figure 1: System Architecture Diagram

This architecture ensures that each service can be developed, tested, and deployed independently while communicating through well-defined REST APIs. The modular design also allows for easy addition of new services in the future

References

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